



PRN : 25070122073

NAME: DHARMIT KUNAL SHAH

BATCH: CSE A-2

Subject: Programming with Java

ASSIGNMENT No : 12

TITLE : Write a Java program to create and use user-defined packages.

Problem Statement

Write a Java program to create and use user-defined packages.

Learning Objectives

To create and use user-defined packages for organizing Java programs.

Theory

Packages are used to organize related classes and interfaces into namespaces. User-defined packages improve code organization, reduce naming conflicts, and enhance reusability. They also support access protection and modular application development. Java packages simplify project management by grouping logically related components together. Large-scale software systems extensively use packages.

Code:

```
import mypack.MyClass;

public class Main {
    Run | Debug
    public static void main (String[] args) {
        MyClass obj = new MyClass();
        obj.display();
    }
}
```

```
package mypack;

public class MyClass {
    public void display () {
        System.out.println(x: "This is a example of a package!");
    }
}
```

Output:

```
PS D:\Java-Questions\Exp12> java Main.java
This is a example of a package!
```

Exercise

1. Create a Java program by developing user-defined packages for a College Management System. Create separate packages for student and faculty details, and display their information using classes from these packages.

```
package Exercise.Ex1.faculty;

public class Faculty {
    int id;
    String name;
    String department;

    public Faculty(int id, String name, String department) {
        this.id = id;
        this.name = name;
        this.department = department;
    }

    public void display() {
        System.out.println(x: "\n----- Faculty Details -----");
        System.out.println("Faculty ID : " + id);
        System.out.println("Name      : " + name);
        System.out.println("Department : " + department);
    }
}

package Exercise.Ex1.student;

public class Student {
    int id;
    String name;
    String course;

    public Student(int id, String name, String course) {
        this.id = id;
        this.name = name;
        this.course = course;
    }

    public void display() {
        System.out.println(x: "----- Student Details -----");
        System.out.println("Student ID : " + id);
        System.out.println("Name      : " + name);
        System.out.println("Course     : " + course);
    }
}
```

```
package Exercise.Ex1;

import Exercise.Ex1.student.Student;
import Exercise.Ex1.faculty.Faculty;

public class Ex1 {
    Run | Debug
    public static void main(String[] args) {
        Student s = new Student(id: 187, name: "Dharmit", course: "Computer Science");
        Faculty f = new Faculty(id: 201, name: "Dr. Raj", department: "CSE");

        s.display();
        f.display();
    }
}
```

OUTPUT

```
PS D:\Java-Questions\Exp12\Exercise\Ex1> java Ex1.java
----- Student Details -----
Student ID : 187
Name      : Dharmit
Course     : Computer Science

----- Faculty Details -----
Faculty ID : 201
Name      : Dr. Raj
Department : CSE
```

2. Create a Java program by developing a user-defined package named library that contains a class to store and display book details such as book ID, title, author, and price. Import the package into the main program and display the book information.

```
package Exercise.Ex2.library;

public class Book {
    int bookId;
    String title;
    String author;
    double price;

    public Book(int bookId, String title, String author, double price) {
        this.bookId = bookId;
        this.title = title;
        this.author = author;
        this.price = price;
    }

    public void display() {
        System.out.println(x: "----- Book Details -----");
        System.out.println("Book ID : " + bookId);
        System.out.println("Title : " + title);
        System.out.println("Author : " + author);
        System.out.println("Price : ₹" + price);
    }
}
```

```
package Exercise.Ex2;
```

```
import Exercise.Ex2.library.Book;
```

```
public class Ex2 {
    public static void main(String[] args) {
        Book b = new Book(158, "Java Programming", "Herbert Schildt", 650.00);

        b.display();
    }
}
```

```
PS D:\Java-Questions\Exp12\Exercise\Ex2> java Ex2.java
----- Book Details -----
Book ID : 158
Title : Java Programming
Author : Herbert Schildt
Price : ₹650.0
```